NAME: ______________________________ **PERIOD:** _______ **DATE:** __________

Six Activations in Twelve Users

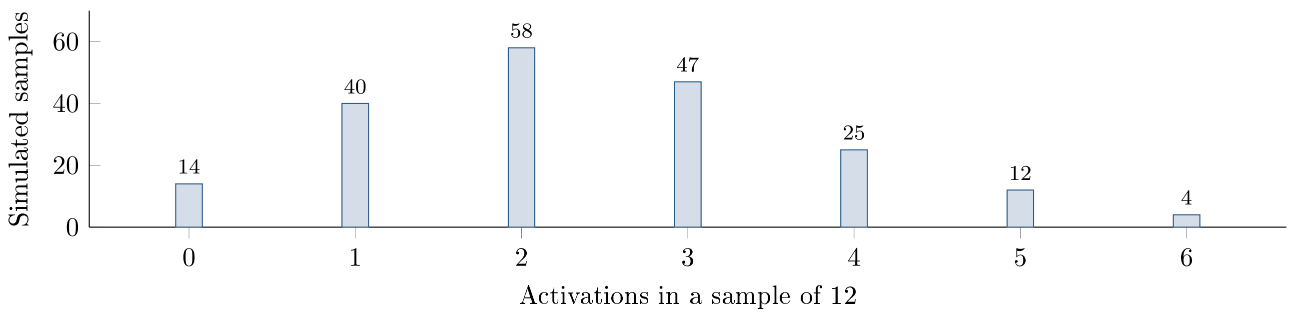
AP Statistics · Unit 3 · Topic 3.3 · B: 2026-12-03 · E: 2026-12-04

[STAR] TODAY'S PROBLEM

Suppose a developer's model says $p_0 = 0.20$ of new app users activate voice commands. An SRS of 12 users includes 6 activations, so $\hat{p} = 0.50$. The chart shows 200 simulated samples of 12 independent users under this model.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



FIRST TAKE — before we discuss (5 min)

Whose method would you trust more, Mara's smooth curve or Jules's repeated trials? Give one reason from the picture.

[BOOK] CHECK THE SHAPE BEFORE USING NORMAL (keep on the page)

Under a proposed proportion p_0, check expected counts $np_0 \geq 10$ and $n(1 - p_0) \geq 10$ before using a normal approximation. A matched simulation estimates the chance probability by the number of simulated results at least as large as the observed result, divided by the total simulations. For this task, a chance probability below 5% counts as convincing evidence against the proposed model. Evidence against a model does not prove a particular cause.

Work (a)–(c) from the rules box:

DOK 1 **(a)** State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

DOK 2 **(b)** Use the chart to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.

DOK 3 **(c)** In 3–4 sentences, choose a method using the simulated tail and either the shape or expected counts. Apply the 5% evidence rule. Explain how reporting 0.005 instead of 0.020 would change a reader's impression.

Frame: I would use ________ because the expected counts are ____________ and the simulated shape is __________.

Frame: The tail estimate ______ means __________________; reporting 0.005 instead would make a reader think ________________.

EXIT — turn this in

One thing the rules box changed about my first take: ______________________